

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Code of Conduct

I **G. Ramya/192572073** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *G. Ramya*

Assessment Tasks

Industry Problem Solving Task

1. An e-commerce company receives clickstream, transaction, and review data from millions of users daily. Propose a big data analytics pipeline and explain how the 5Vs influence architecture design.
2. A healthcare organization collects structured patient records, semi-structured insurance claims, and unstructured medical images. Design an approach to manage and secure this data ecosystem.
3. A telecom company wants to detect churn patterns using high-volume customer interaction data. Identify the big data challenges and recommend suitable analytics and integration strategies.
4. A smart city project generates sensor streams from traffic, air quality, and surveillance systems. Explain how distributed file systems and data integration tools can support large-scale analysis.
5. A financial analytics firm must comply with strict privacy requirements while processing high-velocity transaction data. Design a secure big data solution and justify the controls chosen.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application

Solution Design	25%	Solution is practical, scalable, and well reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

1. E-Commerce Big Data Analytics Pipeline

An e-commerce company receives **clickstream, transaction, and review data** from millions of users.

Proposed pipeline:

Data Sources → Data Ingestion → Storage → Processing → Analytics → Visualization

- **Data sources:** Website clicks, mobile-app activity, purchases, payments, product reviews.
- **Data ingestion:** Apache Kafka or similar tools collect real-time click and transaction events.
- **Storage:** A distributed data lake stores raw and historical data.
- **Processing:** Apache Spark can clean, transform, and analyze large datasets.
- **Analytics:** Machine learning can identify customer behavior, recommend products, and predict sales.
- **Visualization:** Dashboards display sales, customer behavior, and business performance.

Influence of 5Vs:

5V Influence on Architecture

Volume Distributed storage is required for millions of user records.

Velocity Streaming systems are needed to process clicks and transactions quickly.

Variety Different systems are required for structured transactions, clickstream JSON, and text reviews.

Veracity Data validation and cleaning are required to remove incorrect or duplicate data.

Value Analytics and ML convert the collected data into useful business insights.

Conclusion: A combination of **streaming, distributed storage, batch processing, and machine learning** provides a scalable e-commerce analytics solution.

2. Healthcare Data Management and Security

The healthcare organization has three major data types:

- **Structured:** Patient IDs, medical records, laboratory results.

- **Semi-structured:** Insurance claims, XML/JSON records.
- **Unstructured:** X-rays, CT scans, MRI images, medical documents.

Proposed approach:

1. Collect data from hospitals, laboratories, insurance systems, and medical devices.
2. Use APIs and data-integration tools to bring data into a central platform.
3. Store structured records in relational databases.
4. Store large medical images in secure object storage/data lakes.
5. Store claims and other semi-structured information in suitable NoSQL/document storage.
6. Use distributed processing for large-scale analysis.

Security measures:

- Encryption during transmission and storage.
- Role-based access control.
- Strong authentication.
- Patient-data masking or anonymization for analytics.
- Audit logs to monitor access.
- Regular backups and disaster recovery.
- Access should follow the principle of least privilege.

Conclusion: Using different storage systems according to data type, combined with strong security and access controls, provides a scalable and protected healthcare data ecosystem.

3. Telecom Customer Churn Analysis

A telecom company generates huge amounts of customer interaction data.

Data sources:

- Call records.
- SMS and internet usage.
- Customer complaints.
- Recharge and payment history.
- Customer-service interactions.
- Mobile-app activity.

Big data challenges:

- Very high data **volume**.
- Continuous and high-speed data generation.
- Different data formats.
- Duplicate or inaccurate customer information.
- Combining data from multiple telecom systems.
- Protecting customer information.
- Identifying useful patterns from large datasets.

Recommended strategy:

1. Use Kafka or similar streaming tools to collect real-time events.
2. Store historical data in a distributed data lake.
3. Use Spark for large-scale processing.
4. Integrate customer, billing, network, and service data using APIs/ETL tools.
5. Clean and combine customer records.
6. Create features such as usage changes, complaints, payment behavior, and service quality.
7. Apply machine-learning classification models to predict customers likely to leave.
8. Display churn-risk results through dashboards.

Conclusion: Combining **data integration, distributed processing, streaming analytics, and machine learning** can help telecom companies identify churn patterns and take preventive action.

4. Smart City Big Data Architecture

A smart city generates continuous data from:

- Traffic sensors.
- Air-quality sensors.
- Surveillance systems.
- GPS and transportation systems.
- Weather and environmental sensors.

Role of Distributed File Systems:

A distributed file system such as **HDFS** can divide large datasets into blocks and store them across multiple machines.

Advantages:

- Handles very large datasets.
- Provides scalable storage.
- Supports parallel processing.
- Provides fault tolerance through data replication.
- Allows historical sensor data to be analyzed efficiently.

Role of Data Integration Tools:

Tools such as Kafka and ETL/data-integration platforms can:

- Collect data from different sensors.
- Convert different formats into a common structure.
- Filter and clean data.
- Transfer real-time streams to processing systems.
- Combine sensor data with external sources.

Processing architecture:

Sensors → Data Integration → Distributed Storage → Distributed Processing → Analytics → Smart City Dashboard

For example, traffic data can be processed in real time to identify congestion, while historical traffic data can be used to predict future congestion.

Conclusion: Distributed storage provides scalability, while data-integration and streaming tools combine information from different city systems for large-scale analysis.

5. Secure Financial Big Data Solution

A financial analytics firm processes **high-velocity transaction data**, so the system must provide both high performance and strong privacy.

Proposed architecture:

Transactions → Secure Ingestion → Streaming Processing → Encrypted Storage → Analytics → Secure Dashboard

Security controls:

1. **Encryption:** Encrypt transaction data during transmission and while stored.
2. **Tokenization:** Replace sensitive financial identifiers with tokens where possible.
3. **Access control:** Use role-based permissions so employees only access necessary information.

4. **Authentication:** Use strong authentication and multi-factor authentication for sensitive systems.
5. **Data masking:** Hide sensitive information in reports and development environments.
6. **Audit logging:** Record who accessed or modified sensitive data.
7. **Network security:** Use secure connections, firewalls, and network segmentation.
8. **Data retention:** Keep information only for the required period and securely dispose of unnecessary data.
9. **Real-time monitoring:** Detect unusual access and suspicious transactions quickly.
10. **Backup and recovery:** Maintain protected backups to ensure business continuity.

Analytics:

- Stream processing can analyze transactions immediately.
- Machine learning can detect unusual transaction patterns.
- Historical data can be analyzed for fraud trends and financial forecasting.

Conclusion: A secure financial big data system should combine **high-speed streaming, encrypted distributed storage, strict access control, privacy protection, monitoring, and auditing** to meet both analytics and privacy requirements.